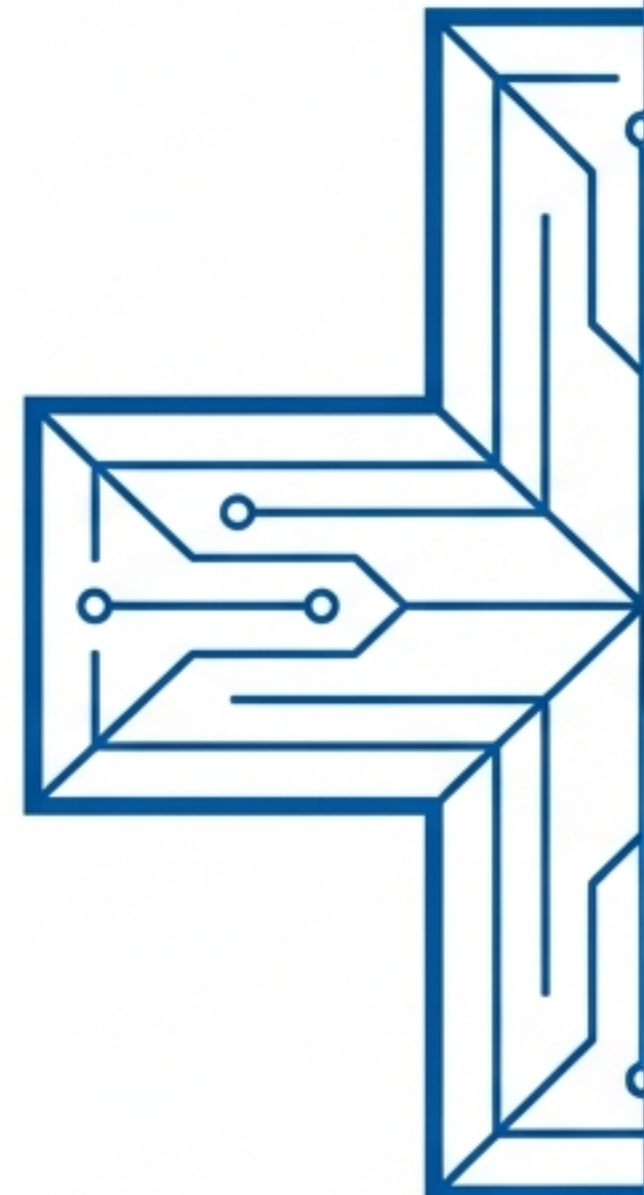


FHIR + AI Agents: Building Clinical Intelligence

From Manual Queries to Autonomous Pipelines | BMI 512 Hackathon



```
import fhir.resources
from ai_agents import ClinicalPipeline

def analyze_patient_data(patient_id):
    # Initialize FHIR client
    client = fhir.resources.FHIRClient(url="https://hapi.fhir.org/baseR4")

    # Retrieve patient record
    patient = client.resources('Patient').get(id=patient_id)

    # Activate AI Agent
    agent = ClinicalPipeline(model="med-palm-2")
    insights = agent.process(patient.resource)

    # Generate clinical intelligence report
    report = agent.generate_report(insights, format="json")

    return report

# Autonomous execution
pipeline_status = analyze_patient_data("P12345")
print(f"Clinical Intelligence Status: {pipeline_status['status']}")
# [END OF STREAM]
```

commit tag
Build: v1.0 // Session: Orientation

The Clinical Scenario: Population Health Management

Mission: Find Type 2 Diabetes patients, retrieve demographics, and flag poor glycemic control.

Patient Profile Card



Target Profile: Type 2 Diabetes Cohort

⚠ HbA1c: 8.2% (High Risk)

Data Fields

Condition: Type 2 Diabetes Mellitus (SNOMED 44054006)

Metric: Glycemic Control

Threshold: > 7.0%



Data Note

Threshold set at >7.0% due to Synthea synthetic data skew (clinical guidelines typically use 7.0-9.0%).

Speaking the Language: FHIR Fundamentals

FHIR is the RESTful API for healthcare. Resources are the building blocks.



Condition

The Diagnosis
(SNOMED 44054006)



Patient

The Person
(Demographics)

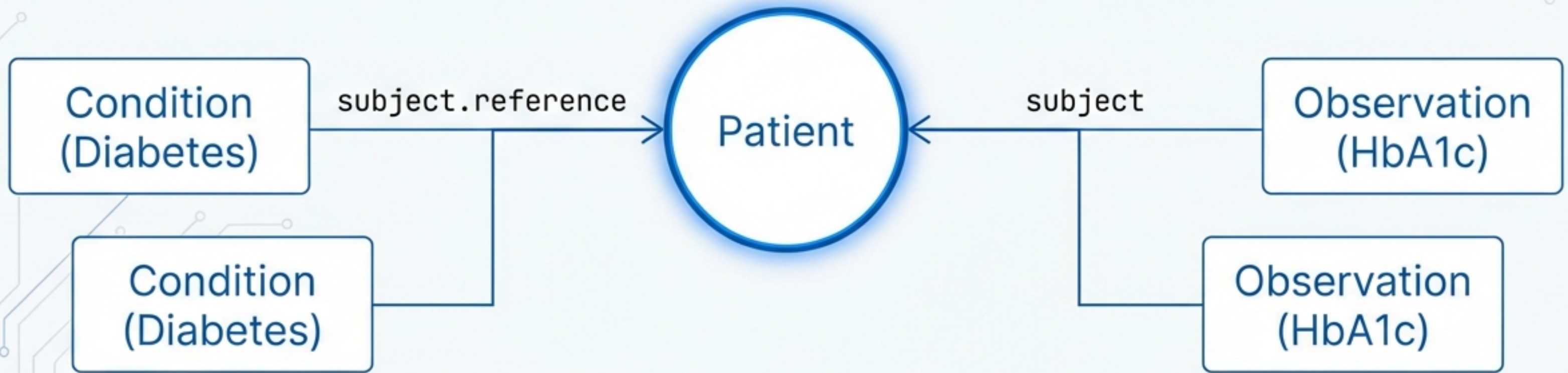


Observation

The Data
(LOINC 4548-4)

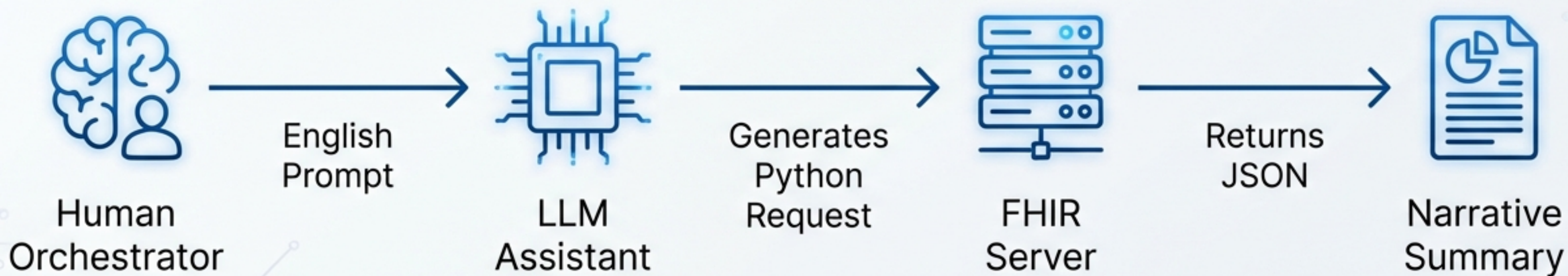
The Knowledge Graph

No single query gives the full answer.
We must chain resources together.



Session 1: The Manual Orchestrator

Human-in-the-Loop Workflow



Key Insight: The LLM never touches the FHIR server directly. YOUR CODE does the querying.

The Pipeline Recipe

The specific algorithmic steps for the Session 1 notebook.

```
pipeline_recipe.fhirstack x

// STEP 1: FIND PATIENTS
query = GET /Condition?code=44054006 (SNOMED: Diabetes)

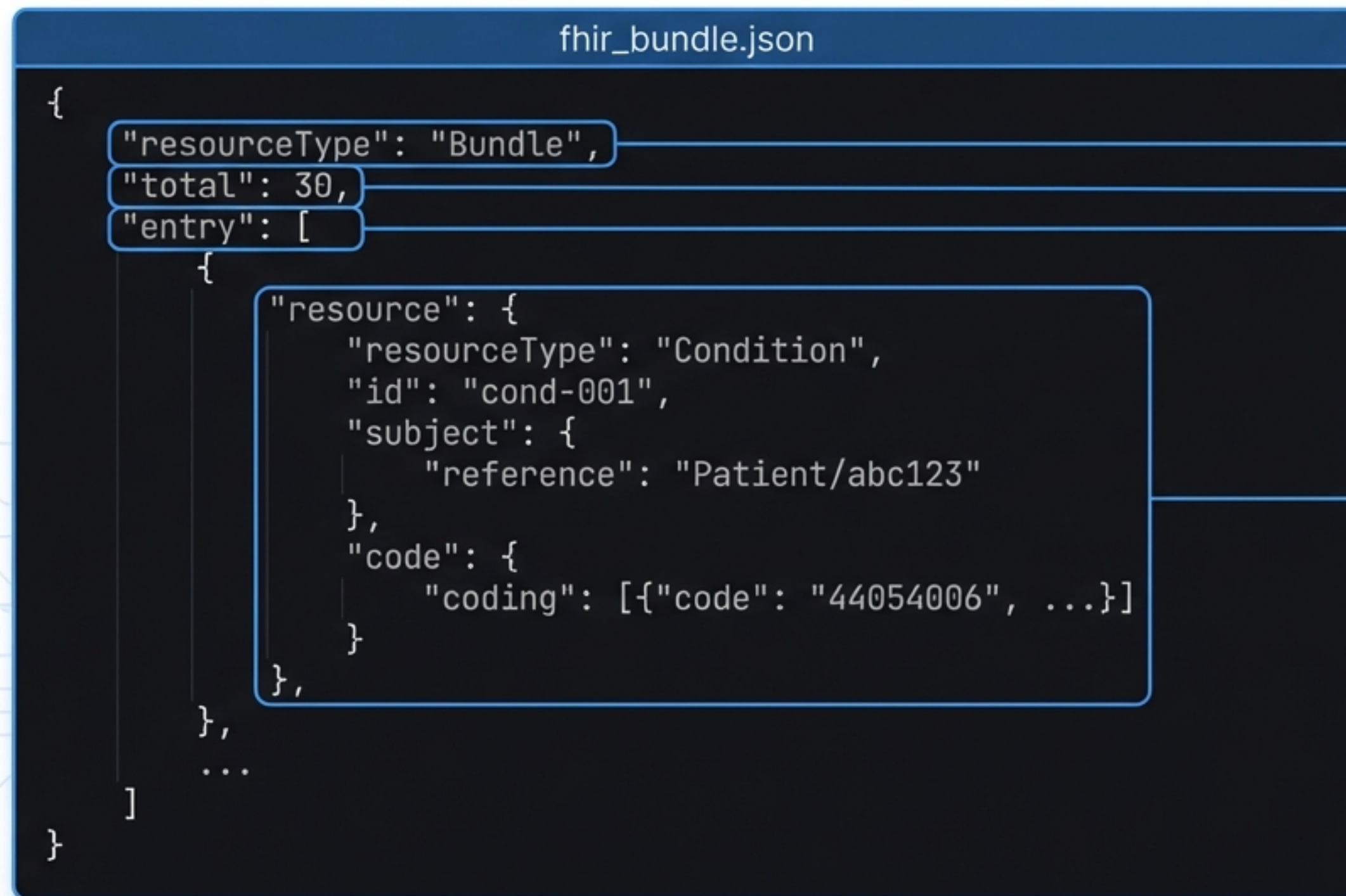
// STEP 2: GET DEMOGRAPHICS
for each condition: retrieve subject.reference -> Patient/{id}

// STEP 3: FIND LAB RESULTS
query = GET /Observation?subject={id}&code=4548-4 (LOINC: HbA1c)

// STEP 4: ANALYZE
if value > 7.0%: flag as 'Poor Control'
```


Anatomy of a FHIR Bundle

Structured data handling, not magic.



The Container

Total Matches

Array of Results

Clinical Data
(Condition/Patient)

Session 2: The Paradigm Shift

Paradigm Shift: Tool Use

Session 1: Manual



Human Driver: You decide the sequence.

Session 2: Agentic

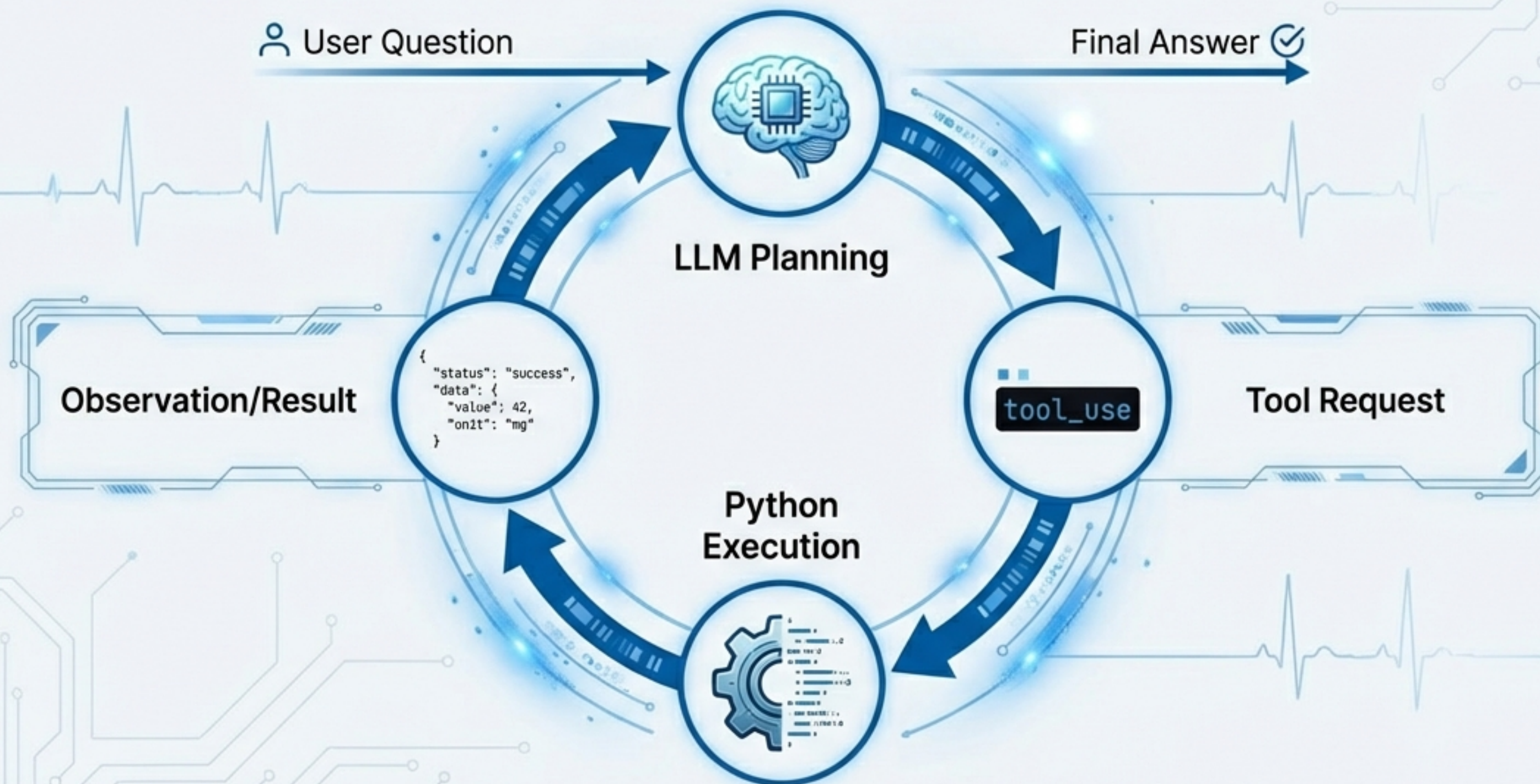


AI Driver: The Agent decides the sequence.

Definition: “**Tool Use**” means the **LLM requests a function call**, and your code executes it.

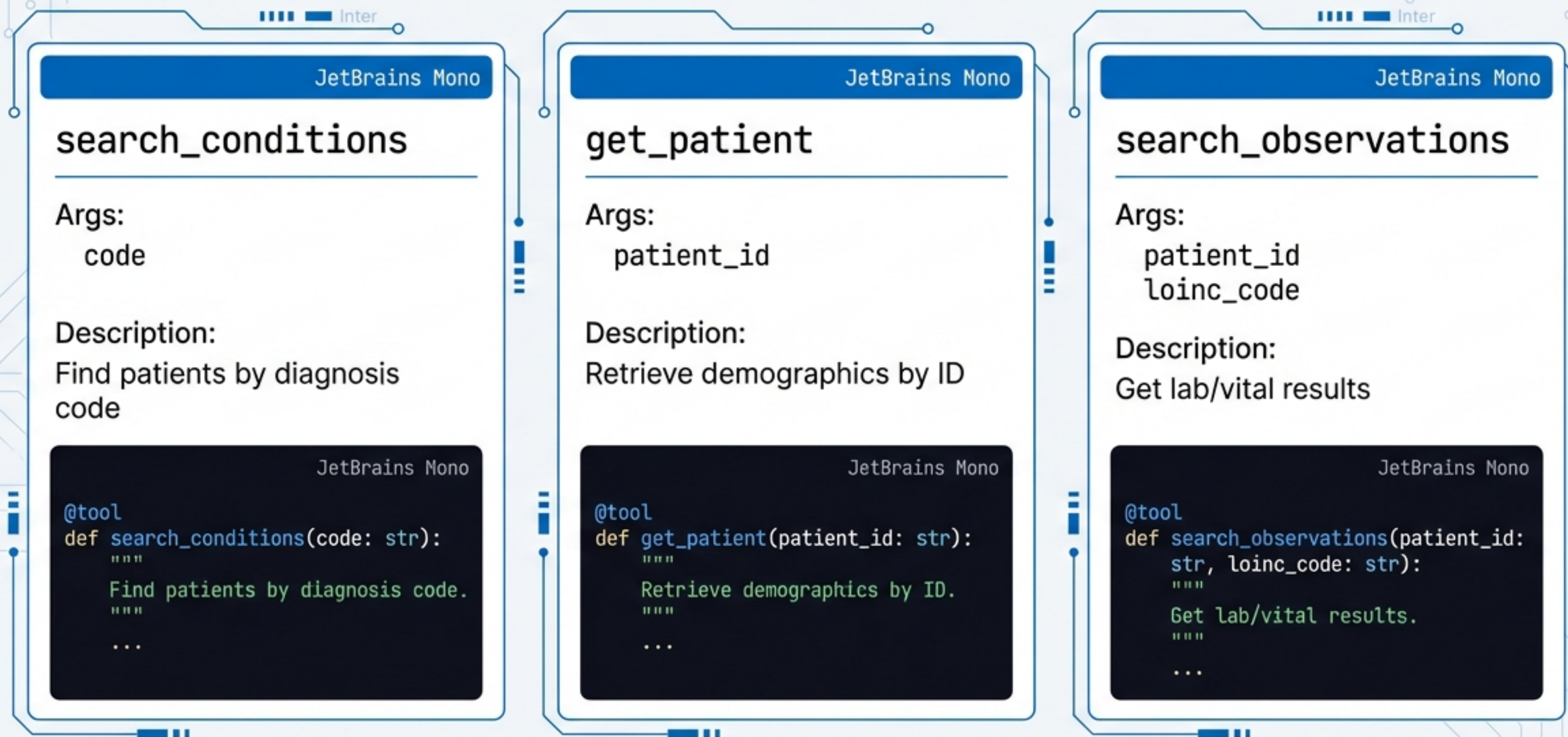
The Agent Loop Architecture

The LLM plans. Code executes. The loop cycles until the mission is complete.



Defining the Tools

The LLM reads the description and schema, not the Python code.



Reading the Trace (Debug Mode)

Validating that the agent is actually performing the retrieval steps.

DEBUG CONSOLE

```
> AGENT START: "Find diabetic patients with high HbA1c"  
[STEP 1] calling tool: search_conditions(code='44054006')  
      <-- found 30 conditions  
[STEP 2] calling tool: get_patient(id='abc-123')  
      <-- found patient: Jane Doe, 1965  
[STEP 3] calling tool: search_observations(id='abc-123', code='4548-4')  
      <-- found value: 8.2 %  
> SYNTHESIS: "Patient Jane Doe has an HbA1c of 8.2%..."
```


Session 3: Expanding the Toolkit

New Capabilities. New Failure Modes.



SYSTEM EXPANSION: ACTIVE. INTEGRATION IN PROGRESS. [STATUS: OPTIMIZED]

Failure Modes & Safety

The goal of Session 3 is to break the agent.

Hallucination



Inventing SNOMED codes that don't exist.

Wrong Tool



Calling search_conditions for lab results.

Infinite Loop



Hitting max_steps without an answer.

The Guardrails Reliability = System Prompts (Strategy) + Tool Descriptions (Instructions)

The Architect's Role

Key Takeaways from the Hackathon.

Separation of Concerns

LLM Plans vs.
Code Executes.

Standards as Bridges

SNOMED & LOINC
are the universal
language.

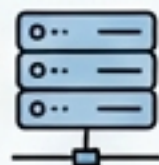
Human as Auditor

From writing
queries to
verifying logic.

[SYSTEM STATUS: INTEGRATION OPTIMIZED]

Mission Resources

Keep this slide open during the hackathon.



Server

```
https://launch.smart  
healthit.org/v/r4/fh  
ir
```



Critical Codes

- Type 2 Diabetes:
SNOMED 44054006
(Warn: Do not use ICD-10)
- HbA1c: LOINC 4548-4



Reminder

Save your Session 1
Notebook—it is your
ground truth reference.

SYSTEM STATUS: INTEGRATION OPTIMIZED